

CS 432/532 - Computer and Network Security

Spring 2026

This is a 3-credit course that focuses on security applications and cryptographic protocols. An overview of cryptography will be given in the first couple of weeks.

This is a code-shared course with both graduate (CS532) and undergraduate (CS432) codes. However, the topics are undergraduate topics and the exams/homework assignments will be designed with the assumption of being an undergraduate course. Graduate students (CS532 takers) will have relatively more load as compared to undergraduate students (CS432 takers). Project grouping rules will be different for CS432 and 532. Graduate students (CS532) will have an extra (but small) takehome exam. Midterm and final exams, and homework assignments will be common, but the letter grades will be determined separately.

Catalogue Data: Overview of Cryptography, Identification and Authentication, Access Control, Operating System Security (UNIX and Windows Environments), Key Distribution, TCP/IP Security, IPSec, DNSSEC, WWW Security, SSL and TLS, E-mail Security, PKI and certificate systems, Firewalls, Intrusion Detection

Prerequisite: Students are expected to come with undergrad level computer networks and basic operating systems background. Moreover, computer-programming expertise is necessary. For CS532, there is no formal prerequisite since it is a graduate course. For CS432, CS408 or EE414 is a prerequisite. However, if you have not taken one of these courses but have a background on Computer Networks, feel free to inquire with the instructor for any possible prerequisite override

Instructor: Albert Levi
FENS 1091, x9563, levi at sabanciuniv edu

Assistants: TBD

Schedule: Lecture: T 14:40 – 15:30 FENS L045 and Th 16:40 – 18:30, FASS G062
Lab: W 14:40 – 16:30, FENS L045

Text book: Cryptography and Network Security, 7th or 8th edition (6th edition OK), William Stallings

References: Computer Security, Dieter Gollmann
Computer Security: Principles and Practice, William Stallings and Lawrie Brown

Tentative Outline

- ❑ Introduction and Basic Concepts (1 week)
- ❑ Overview of Cryptography (2-3 weeks)
 - Symmetric and Asymmetric Cryptography
 - Key agreement
 - Hash functions
- ❑ Authentication and Key Distribution Protocols (1-2 weeks)
- ❑ Kerberos and Password Management (1 week)
- ❑ IP Security and IPSec (2 weeks)
- ❑ Transport Layer Security, SSL/TLS, SSH (1 week)
- ❑ E-mail Security (PGP, S/MIME, Domainkeys, SPF) (2 weeks)
- ❑ PKI and certificate systems, (1 week)
- ❑ Access Control (1 week)
- ❑ Firewalls and Intrusion Detection Systems (1-2 weeks)

Exam Details and Make-up Policy

There will be one midterm and one final exam.

No handheld electronic devices (phones, tablets, smart watches) are allowed in the exam rooms. They have to be left outside. Moreover, you are not allowed to wear any type of watches during the exam. Any violation of these rules will be considered as attempt to cheating.

If you miss the midterm exam with a valid reason, I can arrange compensation that might include the options (picked by me) of oral exam, using final exam instead, some extra questions in the final, an extra exam, or any combination of them. If you miss the finals exam with a valid reason that I accept as well, compensation mechanism will be determined later. Please do not ask questions about the makeup exam details and timing; when I determine them, will let you know.

You can miss **only one of the exams** during the semester (of course with a valid excuse). Definitely, no makeup will be given for the missing makeups whatsoever.

Student responsibilities and loads (tentative)

Common responsibilities and loads (for both CS432 and CS532 students)

- ❑ One midterm and one final exam.
- ❑ There will be 4 (+/- 1) labs. The labs will be dedicated to some practical aspects of the course including programming. Labs will be graded either as in-lab performance or as a separate homework or as after-lab performance. Aside the lab homework assignments, there will be 1-2 lecture related homework assignments. Some homework assignments may require programming.
- ❑ The labs **WILL NOT** be direct application of the lectures, but they will be related to each other. **We DO NOT aim to use labs as recitations to help the students to get higher marks in the exams.**
- ❑ A programming project on a secure networking application. I will decide the number of stages of the project and number of people per group (tentatively 2 people per group) after the semester starts. However, CS532 students should make the project alone. However, CS532 students have an option to propose their own project instead of doing the common one by sending me a proposal within one week after the project will be assigned.

Additional responsibilities and loads for only CS532 students

- ❑ **Takehome exam:** A small takehome exam will be given towards the end of the course. You may consider this takehome exam as a challenging homework as well.

Use of SUCourse and Communications

Announcements will be done via SUCourse that you will also receive as emails. Some announcements may be sent as plain email.

All lecture materials, homework and assignments will be posted at SUCourse. The submissions will also be there unless otherwise stated.

Lecture materials will be posted as PowerPoint file. Each PowerPoint file will be shared after it is entirely covered in class since I can change it even in the last minute.

Tentative Grading and Timing

Midterm Exam	30%	April 15 2026, Wednesday, 14:40-16:30 (lab hour)	week 9
Final Exam	40%	as scheduled by ÖK/SR	
Homework, Lab, Project	30%	deadlines will be determined separately. Individual weights will be determined later. The weight of the project will be between 12-15%.	

CS408 AI Policy

- Course materials (lecture notes, assignments, sample questions, project spec., etc.) are provided only for your personal use. They must not be uploaded, shared, or given, wholly or partially, to any AI tool, service, or platform.
- Students may use AI tools only to reinforce their understanding of course topics (e.g., by asking clarifying questions or exploring related examples).
- Assignments, projects, and other graded work must be completed entirely by the student's own effort. Asking AI tools to directly or indirectly solve or generate answers for these tasks will be considered plagiarism.
- Any violation of these rules will be treated as an academic integrity issue and will be reported to the dean's office for disciplinary investigation.

Use of AI tools will also be considered as plagiarism for any grading item. Plagiarized assignments will not be graded.

For the exams, any type of cheating (including having an electronic device other than calculator in proximity) will be reported to the Dean's Office for disciplinary actions